Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Full Stack Development

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LinkedIn | GitHub | Portfolio

Professional Summary

Full-stack software engineer with 5+ years of experience building SaaS, fintech and e-commerce products, with a focus on real-time, data-heavy systems. Now working on AI engineering: LLM agents, RAG and tool calling in production. Main stack: Python, TypeScript, Nest.js, React.js, Next.js, PostgreSQL, Kafka, AWS and Docker.

Work Experience

Full-Stack AI Engineer (full-time), *OSPHIM* 07/2026 – Present
Aachen, Germany

- Migrated the AI agent from a self-hosted Qwen stack to the Anthropic Claude API, removing the legacy engine.
- Upgraded the main model for chat, reports and handover extraction from Claude Sonnet 4.6 to 5.5, cutting cost per request by about 13%; cheap tasks run on Haiku 4.5.
- Built a FastAPI backend and 12 agent tools so operators can log defects, write handovers and edit assets from chat, with human confirmation on every write.
- Designed the agent to report only what it can verify, e.g. "detection not configured" instead of a false "no anomaly", and kept user PII out of the model.
- Built evals on real production questions to catch regressions before each release.
- Built the shift-handover module end to end, from shift plans and rostering to chat and voice entries, used across 3 plants and 40+ shifts per week.
- Shipped 56 merged PRs across NestJS, SvelteKit 5, PostgreSQL and Redis, with 250+ unit, integration and E2E tests.
- Made real-time machine-data views up to 160x faster with roll-up tables over a 1.4-billion-row table, from 1,150 ms to 16 ms.

Full-Stack Engineer (part-time), *OSPHIM* 01/2026 – 07/2026
Aachen, Germany

- Built the first AI assistant on self-hosted Qwen models, 35B for chat and 1.5B for routing, using LlamaIndex ReAct agents and a Neo4j knowledge graph.
- Built voice handovers on self-hosted Whisper, matching spoken machine names to assets.
- Rebuilt the production logbook as a real-time feed with NestJS, WebSockets and Svelte, linking 5+ entity types per entry.
- Migrated all logbook entries to JSONB with Drizzle ORM, with zero downtime.
- Extended experiments from single- to multi-cavity molds: one setup for a 16-cavity mold instead of 16.

Software Engineer, *MobileLive* 05/2024 – 10/2025
MonieCorp (Fintech Product) Remote

- Developed responsive and modular UI components in React + TypeScript + ShadCN, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant React components for Admin panel using Cursor IDE for rapid development, cutting client feedback cycles from 3 days to 24 hours.
- Built multi-step React form for admins to configure tiered currency margins with validation; streamlined setup from 20+ minutes to <4 minutes per currency pair.

Staples Canada (MobileLive Client)

- Implemented Mixpanel tracking for 100,000+ users, capturing product views, add-to-cart, and checkout events, enabling business team to optimize flows and boost conversions by 12%.
- Built custom React features using Canva API for Staples Online Print Shop, enabling dynamic product personalization and increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

Associate Software Engineer, *Code-Huddle* 08/2022 – 05/2024
GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan

- Built Nest.js APIs with JWT RBAC, handling 10K DAU at <200ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by approximately 30% using FFMPEG compression and optimized AWS CloudFront edge delivery.
- Automated end-to-end CI/CD pipeline with Github Actions, AWS CodePipeline, cutting deployment time by 40%.

Ordinami: QR Ordering Web-App (Italian Startup MVP)

- Implemented Firebase Auth + Firestore backend, supporting 100+ daily orders.
- Built an admin-panel shadcn/ui + Tailwind for menu creation, reducing manual setup time for owners by 60%.

Product Development Intern, *Code-Huddle* 04/2022 – 06/2022
Islamabad, Pakistan

- Assisted in frontend (React.js) and backend (NestJS) development, contributing to the design and implementation of scalable solutions.
- Supported AWS operations and collaborated on ongoing projects, gaining exposure to the full product development lifecycle.

Education

Bachelor of Science in Software Engineering, *COMSATS University* 2018 – 2022
Islamabad, Pakistan

Skills and Core Competencies

AI & LLM Engineering
Python | FastAPI | SQLAlchemy | Claude API | Qwen / Ollama | LlamaIndex | LLM Agents | Tool Calling | RAG | Evals | Whisper | Neo4j

Frontend Development
React.js | Next.js | SvelteKit | Context API | Tailwind CSS | HTML5 | CSS3 | Shopify (Liquid Templating) | ShadCN UI | Redux and Redux Toolkit | RxJs

Software Development & Engineering
Full-Stack Web Applications | RESTful APIs | Microservices Architecture | Secure Authentication Systems | Payment Integrations (Stripe) | CI/CD Automation | SaaS Product Engineering

Cloud, DevOps & Containerization
AWS (ECS, ECR, EC2, RDS, S3, CloudFront, CodePipeline, CloudWatch, SNS) | Azure | Docker

Backend Development
JavaScript | TypeScript | Node.js | Express.js | Nest.js | TypeORM | Prisma | Drizzle ORM | Kafka | Redis

Databases & Data Management
MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling

Notable Freelance Projects

Pendfunds.com (M.V.P.), *Contract Management & Mediation Software* 01/2024 – 09/2024

- Developed full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in Nest.js, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging, and production environments using AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented secure payload encryption/decryption using PGP with RSA-256, ensuring sensitive transaction data remained fully protected.